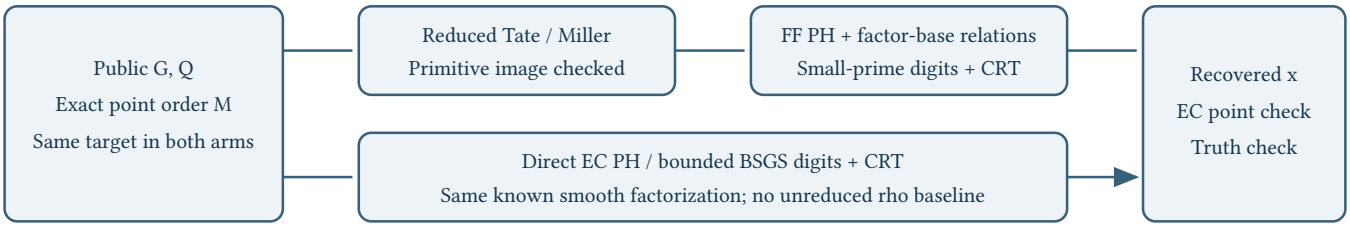
Original C: complete selected point orders

Run source. Started 2026-10-17T19:09:32.476381+00:00. Host: INTEL(R) XEON(R) PLATINUM 8573C, x86_64, kernel 6.18.55. Population: original source model only. Primary observations; replays are kept separately.



Native accounting: 4 method records; 2 matched verified targets. Point orders M range from 545986327049404416 to 545986327049404416. Both arms recover modulo the complete verified M. A selected point need not generate the full curve group; maximality is not asserted.

The field is p=1637958981148213249, and N=p-1=2^26*3^20*7. Every selected point order is smooth over 2,3,7. Its large numeric value is not a large-prime security level. Both arms exploit the known factorization. Solver arguments contain the public curve, G, Q and M, never scalar truth. The runner verifies the recovered point and the separate public known answer.

The finite-field route varies the first argument of reduced Tate/Miller, checks exact prime-power image order, then performs multiplicative PH. Initially unknown projected small-prime base logs are recovered by checked one-/two-factor relations and modular Gaussian elimination. No generic-log fallback is used. This is a smooth-order diagnostic, not NFS/integer-smoothness index calculus, a large-k backend or a new asymptotic result.

Recorded finite-field setup equations/queries: 14 / 14. Target digit equations/queries: 92 / 92. Identity and dependent setup equations remain counted; per-component ranks, matrix work and pair tests are retained. Tiny prime digit groups explain the dense relation coverage; these yields do not describe a large-order oracle.

Fixture construction, exact order finding and known-target generation are recorded separately, outside both solver intervals. Cold solver work includes input-order checks, projections, method setup, auxiliary searches, pairing, relations/matrix work, digit recovery, CRT and final EC point verification. Raw end-to-end observations remain in the native CSV, with min/max/mean and n. No admitted runtime comparison is made: one round, no A/A, PSI unavailable. CPU 4 was reserved; wrapper observed contended=false. Replays do not contribute to primary means. The full-prefix replay is explicitly contended.

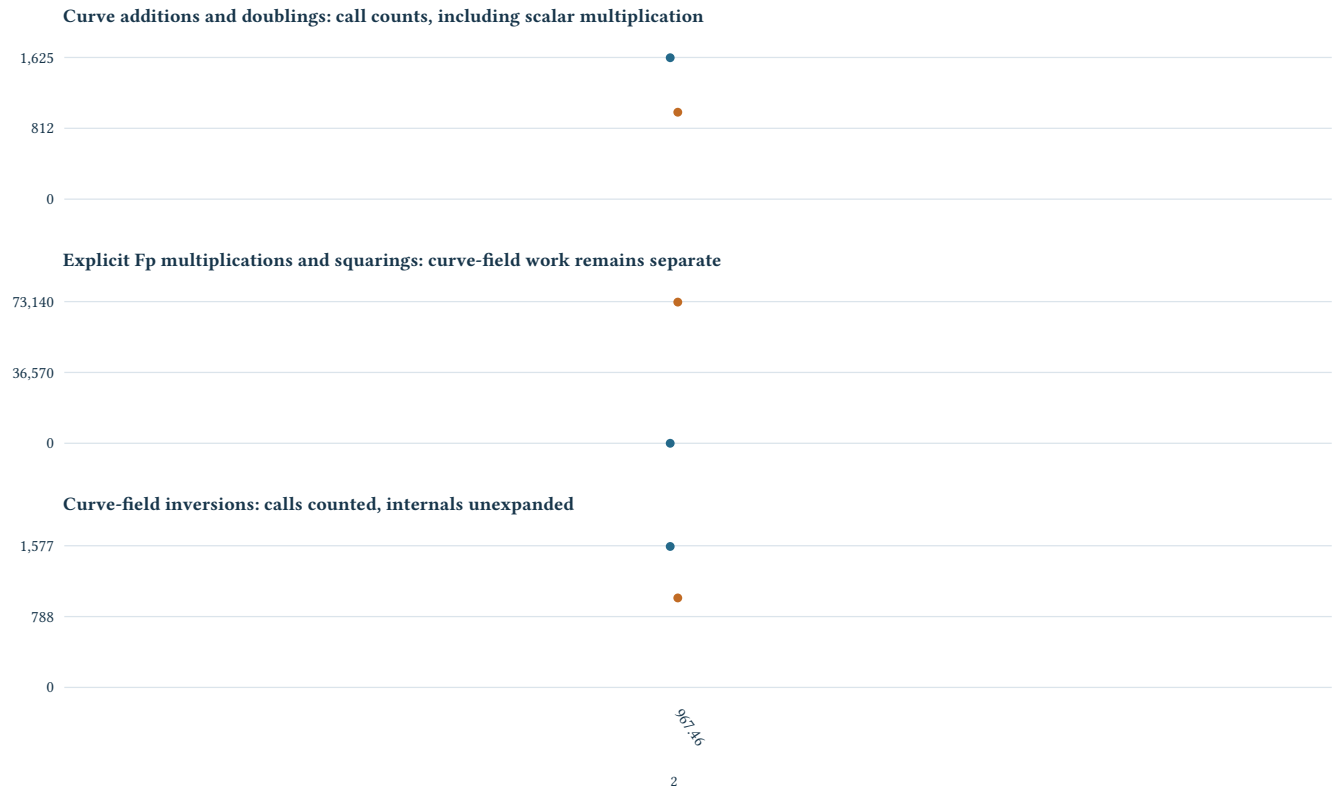
The immutable C prefix contains 1,869 vertices and 4,903 replayed maps; 1,024 vertices are expanded. Local labels certify 517 vertices / 24 observed strata and keep 1,352 unknown. Original full C remains 1,934,901,248 vertices and 165 strata. This does not complete original A–E, the full eight-target / five-round matrix, transported-target comparisons, representation registration or independent arithmetic replay.

Raw records, fixtures, scalar truth, native CSV/statistics, isolation journals, exact binaries and both source/replay populations accompany the delivery. Native implementation is source-pinned at d954e80d9; linked-library source 2bf7700b7. Source hash `bc90c72be7042e6ff355ed529302f2fcbc48d5e0e4ce15c4bd6dd241cd179234`.

Recorded primitive work across local strata

Original C: full selected-point-order diagnostic — source

Native per-target means and observed min–max, not confidence intervals. Blue: EC control. Orange: finite-field route.



Conductor f in millions, ordinal positions; ? keeps unknown labels. Bottom numbers: target observations per arm.

Point orders vary and are certified; these samples do not isolate a causal conductor effect or establish a priced total.

Arm	Metric	n	Min	Mean	Max
EC PH / BSGS	Curve add + double	2	1,619	1,622.00	1,625
EC PH / BSGS	Explicit Fp mul + square	2	0	0.00	0
EC PH / BSGS	Curve-field inverse	2	1,561	1,569.00	1,577
Tate + FF PH relations	Curve add + double	2	993	997.00	1,001
Tate + FF PH relations	Explicit Fp mul + square	2	72,622	72,881.00	73,140
Tate + FF PH relations	Curve-field inverse	2	992	996.00	1,000

These are native descriptive summaries. Curve additions/doublings, curve-field inversions and explicit Fp multiplication/square counts are different units; they are not a priced total. EC inversion and square-root internals remain unexpanded. Explicit Fp inversions already contribute to the multiplication / square ledger and must not be priced twice. Allocation, comparisons, scalar integer work and Gaussian inverse details remain unpriced with counters where available. All per-phase counters and present-value n are in the native CSV. Observed ranges are not confidence intervals. Unknown strata remain separate. Source and replay plots may match because every non-clock field reproduces; they are not extra independent samples or an independent arithmetic proof.